Micropore influence on TENG charge retention

"How do micropores affect charge retention time in TENGs" *Generated by LeapSpace*

Micropores generally **increase charge retention time in TENGs** by creating more internal trap sites and localized polarization regions that hold triboelectric charges longer, but **too many micropores can shorten retention indirectly** if they make the structure too thin, too open, or leak-prone [89] [95] [98] [99] [100] [101] [102] [103] [110].

What micropores do to retention time

Micropore feature	Effect on charge retention time	Why it happens
More micropore surfaces and edges	Longer retention	Micropores add trap-rich internal surfaces where charges can be held instead of quickly recombining or leaking away [89] [98] [100] [102] [110].
Higher internal polarization	Longer retention	Microporous structures create interfacial polarization and local dipole regions that stabilize stored charge [90] [95] [99] [101].
Hierarchical porous networks	Longest retention when balanced	A connected porous network can trap charge while still preserving enough mechanical integrity for repeated cycling [100] [101] [103].
Excessive microporosity	Shorter effective retention	If pores become too dense or too open, sidewalls weaken, charge leakage rises, and output drops [99] [104] [110].

Mechanism in plain terms

Micropores slow charge loss by **capturing charges at pore walls, defect-rich surfaces, and internal interfaces**, which reduces recombination and diffusion to the electrode [89] [90] [100] [102] [110]. Several studies explicitly link microporous or porous structures to **reduced charge decay** and **more persistent surface potential** during cycling [100] [101] [102]. In one device, increasing micropore arrays improved output up to an optimum, but too many pores caused the film to lose elastic support and the electrical output to fall, showing that retention benefits are **not unlimited** [110].

Bottom line

Micropores usually lengthen charge retention time in TENGs, but only up to an optimal porosity level. The best designs use **controlled microporosity** to maximize trap density and interfacial polarization while avoiding structural weakening and charge leakage [100] [101] [102] [103] [110].

Confidence level: **High** — multiple retrieved TENG studies consistently connect microporosity with stronger charge trapping and slower decay, and the main limitation is consistently the same: excess porosity can reverse the benefit.

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